

Agenda

- Inferences Concerning Population Variance
- Inferences Concerning Two Population Variances

Inferences Concerning the true variance of a random variable X

- $X_1, \dots, X_n$: random sample

- Assumptions:

– $X_i \sim N(\mu, \sigma)$ $\longrightarrow$

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$$

Normality is crucial, CLT does not apply here!!!

A new test statistics to check $H_0: \sigma_1^2 = \sigma_0^2 \dots$

Tests about a population Variance and Standard Deviation of a Normal r.v.

$$H_0: \sigma_1^2 = \sigma_0^2$$

$$\text{Test statistics: } \chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

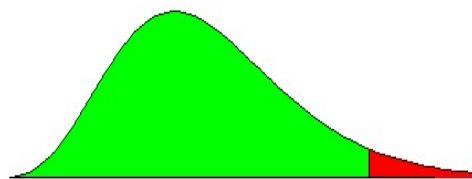
$$H_a: \sigma_1^2 > \sigma_0^2$$

$$H_a: \sigma_1^2 \neq \sigma_0^2$$

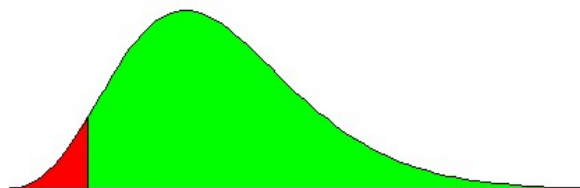
$$H_a: \sigma_1^2 < \sigma_0^2$$

Accept

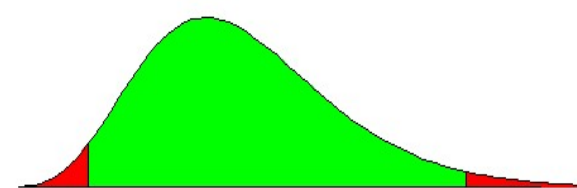
Reject



$$\text{R.R. : } \chi^2 \geq \chi_{\alpha; n-1}^2$$

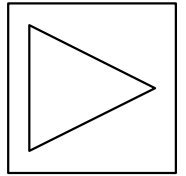


$$\text{R.R. : } \chi^2 \leq \chi_{1-\alpha; n-1}^2$$



$$\begin{aligned} \text{R.R. : } & \chi^2 \leq \chi_{1-\frac{\alpha}{2}; n-1}^2 \\ \text{or } & \chi^2 \geq \chi_{\frac{\alpha}{2}; n-1}^2 \end{aligned}$$

Beverage Filling Machine Example



- Machine fills 12 oz per can
- Known variance: $\sigma^2 = 0.01$
- Machine moved to a new location
- Sample size: $n = 10$ cans

Sample:

12.18, 11.77, 12.09, 12.03, 11.87
11.96, 12.03, 12.36, 12.28, 11.85

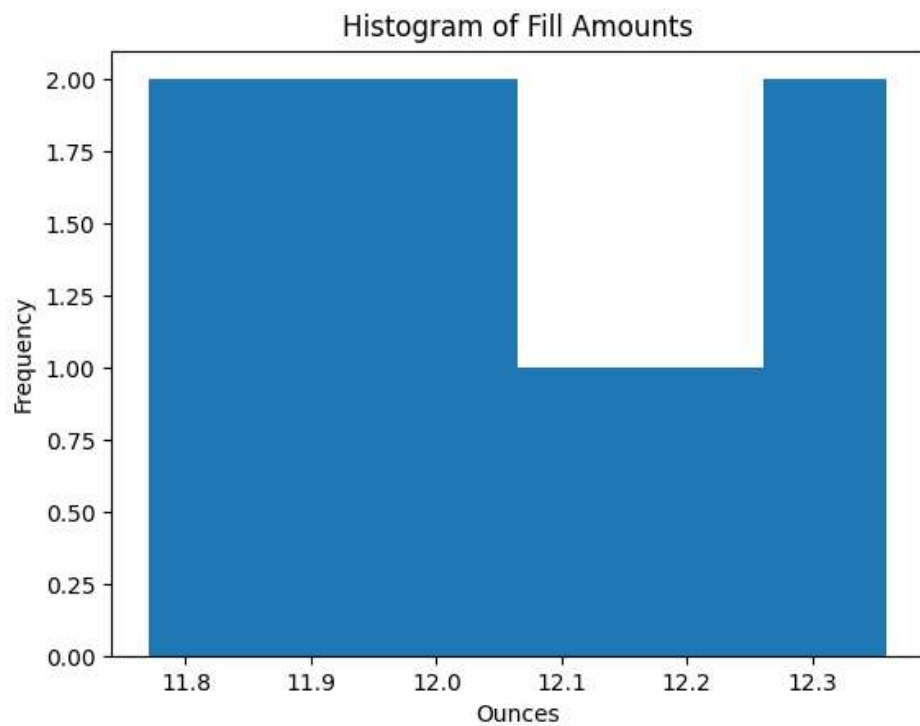
Formulate a hypothesis test:

$$H_0: \sigma^2 = 0.01$$

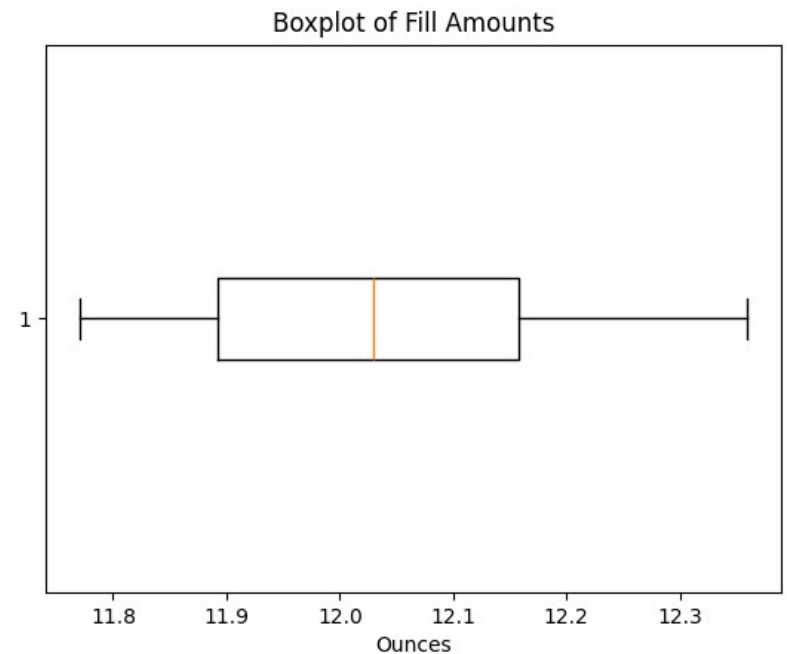
$$H_1: \sigma^2 \neq 0.01$$

Assumptions?

Plots of the sample



Sample skewness: 0.78



Beverage Filling Machine Example

- Machine fills 12 oz per can
- Known variance: $\sigma^2 = 0.01$
- Machine moved to a new location
- Sample size: $n = 10$ cans

Normal sample:

12.18, 11.77, 12.09, 12.03, 11.87
11.96, 12.03, 12.36, 12.28, 11.85

$$S^2 = 0.0363$$

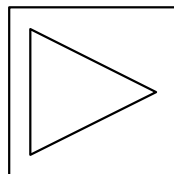
Formulate a hypothesis test:

$$H_0: \sigma^2 = 0.01$$

$$H_1: \sigma^2 \neq 0.01$$

Test Statistic:

$$\chi^2 = \frac{(n-1)S^2}{0.01} = 32.63$$



Reject H_0 at $\alpha=1\%$.

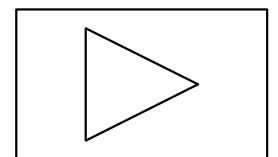
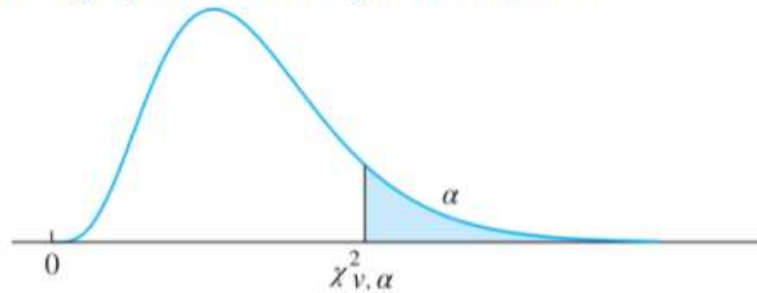
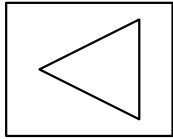


TABLE A.7 Upper percentage points for the χ^2 distribution

$$\chi^2 = 32.63$$



ν	α									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156

Remarks

1. Normality is crucial. AT LEAST AN APPROXIMATE BELL SHAPE IS NEEDED FROM THE HISTOGRAM OF THE DATA.
2. Type-II error, and sample size choice → Software packages...

Inferences Concerning Two Population Variances

- Why do we need these tests?
- For some systems: A CHANGE IN THE VARIANCE OF PROCESS CAN BE IMPORTANT
e.g., variance of contents in a bottle.
- Pooled t-test: CHECK ON THE EQUALITY OF VARIANCES

A new test statistics to check $H_0: \sigma_1^2 = \sigma_2^2 \dots$

Definition: F distribution

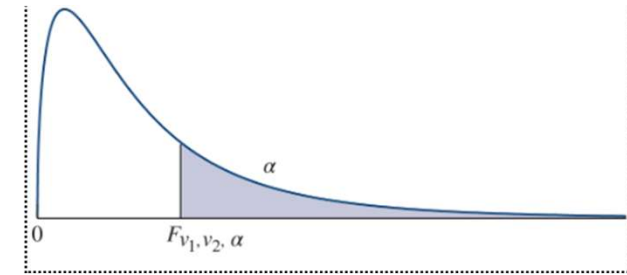
- Let W and Y be χ^2 random variables with u and v degrees of freedom, respectively, then:

$$\frac{W/u}{Y/v} \sim F_{u,v}$$

- Properties: $F_{u,v} = \frac{1}{F_{v,u}}$

$$F_{1-\alpha;u,v} = \frac{1}{F_{\alpha;v,u}}$$

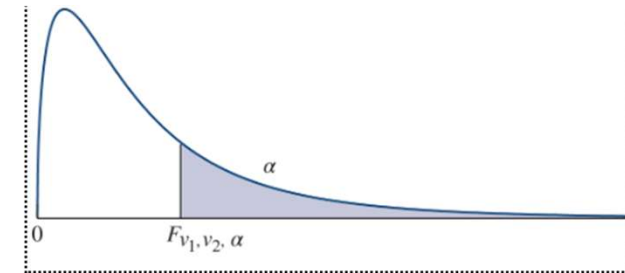
TABLE A.8 Upper percentage points for the F distribution



Example

- $v_1=10, v_2=15,$
- $F_{0.05, 10, 15} = ?$
- $F_{0.95, 15, 10} = ?$

		v_1									
v_2	α	10	12	15	20	25	30	40	50	60	
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30	
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	
	0.010	3.69	3.55	3.41	3.26	3.16	3.10	3.02	2.97	2.93	
	0.001	5.81	5.55	5.27	4.99	4.82	4.70	4.54	4.45	4.39	12

TABLE A.8 Upper percentage points for the F distribution

Example

- $v_1=10, v_2=15,$
- $P\{F_{10,15} > 3.8\} = ?$
- $P\{F_{10,15} > 4.5\} = ?$

		v_1									
v_2	α	10	12	15	20	25	30	40	50	60	
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30	
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	
	0.010	3.69	3.55	3.41	3.26	3.16	3.10	3.02	2.97	2.93	
	0.001	5.81	5.55	5.27	4.99	4.82	4.70	4.54	4.45	4.39	13

$$\frac{W/u}{Y/v} \sim F_{u,v}$$

Theorem

$$X_1, \dots, X_m \sim N(\mu_1, \sigma_1^2) \quad \text{and} \quad Y_1, \dots, Y_n \sim N(\mu_2, \sigma_2^2)$$

$$\Rightarrow F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F_{m-1, n-1}$$

Proof:

$$\frac{(m-1)S_1^2}{\sigma_1^2} = \frac{\sum (X_i - \bar{X})^2}{\sigma_1^2} \sim \chi_{m-1}^2$$

Corollary

$$\text{If } \sigma_1^2 = \sigma_2^2, \quad \frac{S_1^2}{S_2^2} \sim F_{m-1, n-1}.$$

Tests about the equality of two population variances (or standard deviations) of two Normal random variables

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$\text{Test statistics: } F = \frac{S_1^2}{S_2^2}$$

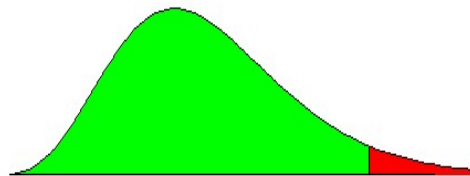
$$H_a: \sigma_1^2 > \sigma_2^2$$

$$H_a: \sigma_1^2 \neq \sigma_2^2$$

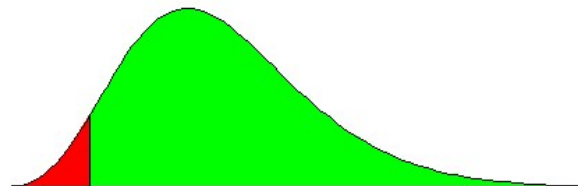
$$H_a: \sigma_1^2 < \sigma_2^2$$

Accept

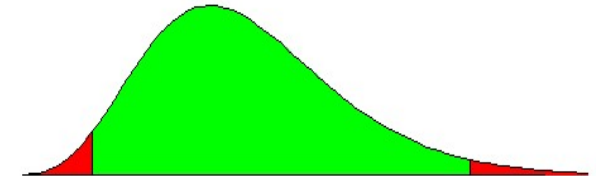
Reject



$$\text{R.R. : } f \geq F_{\alpha; m-1, n-1}$$



$$\text{R.R. : } f \leq F_{1-\alpha; m-1, n-1}$$



$$\begin{aligned} \text{R.R. : } f &\geq F_{\alpha/2; m-1, n-1} \\ \text{or } f &\leq F_{1-\alpha/2; m-1, n-1} \end{aligned}$$

Example

- 12 oz bottles by a new & an old machine

Data

new: $s_1 = 0.1092$ $m = 15$

old: $s_2 = 0.12$ $n = 16$

Hypotheses Test?

$$H_0: \sigma_1^2 = \sigma_2^2$$

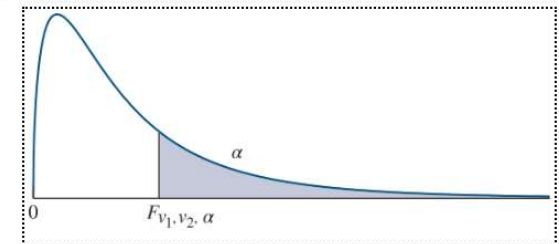
$$H_a: \sigma_1^2 < \sigma_2^2$$

For $\alpha = 0.01, 0.10$ and $\alpha=0.05$?

- Critical value? $F_{14,15,0.95}=?$

Critical value?

TABLE A.8 Upper percentage points for the F distribution



$$F_{14,15, 0.95}=?$$

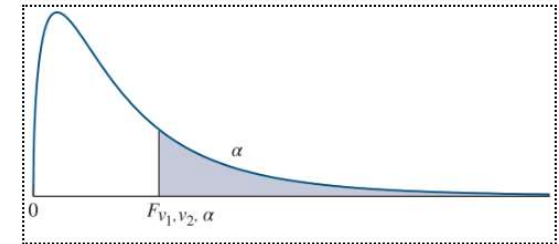
		v_1									Page 834
v_2	α	10	12	15	20	25	30	40	50	60	
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	
	0.001	6.80	6.52	6.23	5.92	5.75	5.62	5.47	5.37	5.30	
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	

Critical value?

$$F_{14,15,0.95}=?$$

$$F_{14,15,0.95} = \frac{1}{F_{15,14,0.05}} = \frac{1}{2.46} = 0.407$$

TABLE A.8 Upper percentage points for the F distribution



		v_1									Page 834
v_2	α	10	12	15	20	25	30	40	50	60	
10	0.100	2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	
	0.050	2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	
	0.010	4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	
	0.001	8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	
11	0.100	2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	
	0.050	2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	
	0.010	4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	
	0.001	7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	
12	0.100	2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	
	0.050	2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	
	0.010	4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	
	0.001	7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	
13	0.100	2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	
	0.050	2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	
	0.010	4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	
	0.001	6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30	
14	0.100	2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	
	0.050	2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	
	0.010	3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	
	0.001	6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	
15	0.100	2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	
	0.050	2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	
	0.010	3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	
	0.001	6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	
16	0.100	2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	
	0.050	2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	

Example

- 12 oz bottles by a new & an old machine

Data

new: $s_1 = 0.1092$ $m = 15$

old: $s_2 = 0.12$ $n = 16$

Hypotheses Test?

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_a: \sigma_1^2 < \sigma_2^2$$

For $\alpha = 0.01, 0.10$ and $\alpha=0.05$?

- Critical value?
- Test statistics, F? $F = \frac{s_1^2}{s_2^2} = 0.8281$
- p-value? P-value > 0.10

$$F_{14,15, 0.99} = 0.273$$

$$F_{14,15, 0.95} = 0.407$$

$$F_{14,15, 0.90} = 0.498$$

Remarks

1. Normality is crucial. AT LEAST AN APPROXIMATE BELL SHAPE IS NEEDED FROM THE HISTOGRAM OF THE DATA.
2. Tests perform best when $m=n$. If m and n are very different and any doubt about normality DO NOT USE THE TESTS.
3. TESTS ARE NOT POWERFUL : H_0 will not be rejected fairly often even if H_0 is not true. Higher α can be a solution, α can be as high as 0.2 for these tests.
4. Type-II error, and sample size choice → Software packages...